SIEU TRAN

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RESEARCH INTERESTS

Self-supervised and structured representation learning for dynamic environments, with a focus on object-centric learning, video understanding, world models, and robot perception. I am particularly interested in learning persistent and hierarchical latent representations that support prediction, reasoning, and long-horizon decision making under partial observability.

EDUCATION

University of Arkansas (UARK)
Department of Electrical Engineering and Computer Science (EECS)
Master of Sciences in Computer Science CGPA: 4.00/4.00 Expected Graduation: 2027

Vietnam National University Ho Chi Minh City (VNU-HCMC)
University of Information Technology (UIT)
Bachelor of Sciences in Computer Science GPA: 9.24/10.0 Graduation Rank: 1st - Valedictorian

EXPERIENCE

University of Arkansas - AICV Lab 2026 – Present

- Proposed a **fully self-supervised** object-centric video framework that decouples per-frame appearance from temporally persistent identity into two slot states, resolving the objective conflict between reconstruction and temporal consistency; evaluated on MOVi-C, MOVi-D, and YouTube-VIS (*DSSA*, ACM MM '26, co-first author).
- Designed a slot-lifecycle mechanism that learns a per-slot, per-frame activation score **without visibility supervision**, preserving latent state for absent or fully occluded objects and preventing state drift and decoder interference (*TSA*, under review, co-first author).
- Built a hierarchical **vision-language-action** framework combining a persistent semantic-graph state with a synthesized code-based planner for non-Markovian long-horizon robot manipulation under partial observability (*CodeGraphVLP*, under review, co-first author).

Day One Company - AI Lab 2024 – 2025

- Proposed a **self-supervised** pipeline enabling OCR models to adapt to new customer domains without annotation effort, reducing onboarding cost by 45%.
- Built information extraction systems generalizing across a wide variety of unseen receipt layouts; raised end-to-end automation from 60% to 90% and reached 99% accuracy on fraudulent-field detection.
- Led research & development, deployment, and maintenance of these AI products in production, including the MLOps pipeline and monitoring system.

UIT - AI Lab 2021 – 2024

- Designed region-aware pose encodings and **signer-invariant** learning strategies that bridge domain gaps across individuals for signer-independent sign language recognition (2 papers, ICCV Workshop 2025; 2nd place, MSLR Challenge).
- Built **multimodal fusion** pipelines combining vision-language embeddings (CLIP, BLIP), OCR, and ASR with temporal modeling for retrieval over 300+ hours of video (SOICT 2023, first author).
- Developed object detection robust to severe class imbalance and adverse visual conditions in real-world traffic footage (CVPR Workshop; 1st place, 8th AI City Challenge).

HONORS AND AWARDS

Graduate Assistantship Award, UARK	2026
Ranked 2nd, Multimodal Sign Language Recognition Challenge, ICCV Workshop	2025
Winner, 8th AI City Challenge, CVPR Workshop	2024
Valedictorian, Bachelor's degree, UIT	2024
Second Prize, Central-Level Science and Technology Awards	2024
2nd Runner Up, FPT AI Challenge	2023
Third Prize, Ho Chi Minh AI Challenge	2023
BOSCH Activator Scholarship in Ambassador	2023
Second Prize, UIT AI Challenge	2022
UIT Scholarships for Outstanding Students	2022–2024

PUBLICATIONS

[1] Sieu Tran, Duc Nguyen Minh, Hao Vo, Khoa Vo, and Ngan Le. Dual-State Slot Attention: Decoupling Appearance and Identity for Video Object-Centric Learning. Proceedings of the 34th ACM International Conference on Multimedia (MM '26), November 10–14, 2026, Brazil. (accepted)

[2] Duc Nguyen Minh, Sieu Tran, Hao Vo, Khoa Vo, Duy Minh Ho Nguyen, Nghi D. Q. Bui, Anh Nguyen, Long Mai and Ngan Le. TSA: Temporal Slot Activation for Persistent Object-Centric Video Representationg. *arXiv preprint*, 2026. (under review)

[3] Khoa Vo, Sieu Tran, Taisei Hanyu, Yuki Ikebe, Duy Nguyen, Nghi D. Q. Bui, Minh Vu, Anthony Gunderman, Chase Rainwater, Anh Nguyen, Ngan Le. CodeGraphVLP: Code-as-Planner Meets Semantic-Graph State for Non-Markovian Vision-Language-Action Models. *arXiv preprint*, 2026. (under review)

[4] Sieu Tran, Duc Nguyen Minh, Truong Nguyen Thanh, Hao Vo, Thanh Duc Ngo, Tien Do, Khiem Le, Duy-Dinh Le. Region-Aware Pose Modeling and Permutation Decoding for Signer-Independent Sign Language Recognition. In ICCV Workshop, Honolulu, Hawai'i, USA, 2025.

[5] Sieu Tran, Duc Nguyen Minh, Truong Nguyen Thanh, Hao Vo, Thanh Duc Ngo, Tien Do, Khiem Le, Duy-Dinh Le. Generalizable Sign Language Recognition via Local Temporal Convolutions and Region-Aware Pose Encoding. In ICCV Workshop, Honolulu, Hawai'i, USA, 2025.

[6] Hao Vo, Sieu Tran, Duc Minh Nguyen, Thua Nguyen, Tien Do, Duy-Dinh Le, Thanh Duc Ngo. Robust Motorcycle Helmet Detection in Real-World Scenarios: Using Co-DETR and Minority Class Enhancement. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops(pp. 7163-7171).

[7] Sieu Tran, Duc Minh Nguyen, Triet Huynh Minh Nguyen, Danh Phuc Ngo, Thu Minh Nguyen, Hao Vo, Khiem Le, Tien Do, Thanh Duc Ngo. Diverse Search Methods and Multi-Modal Fusion for High-Performance Video Retrieval. In Proceedings of the 12th International Symposium on Information and Communication Technology (pp. 997-1002).